

Obstructions to $\mathbb{Q}$ -Acyclic Universal Covers for Lattices with Involutions

1 Main statements

Theorem 1.1 (Integrality forced by a $\mathbb{Q}$ -acyclic universal cover). *Let M be a closed connected n -manifold (or more generally, a finite connected CW complex) with fundamental group $\pi_1(M) \cong \Gamma$. Assume the universal cover $\widetilde{M}$ is $\mathbb{Q}$ -acyclic, i.e. $\widetilde{H}_i(\widetilde{M}; \mathbb{Q}) = 0$ for all i . Then Γ admits a finite free resolution of the trivial $\mathbb{Q}\Gamma$ -module $\mathbb{Q}$ by finitely generated free $\mathbb{Q}\Gamma$ -modules. In particular, the $\mathbb{Q}$ -Euler characteristic of Γ is defined and satisfies*

$$\chi(\Gamma) = \chi(M) \in \mathbb{Z}.$$

Theorem 1.2 (Existence of a uniform lattice with an involution and half-integral Euler characteristic). *There exists a torsion-free cocompact lattice $\Pi < PU(2, 1)$ such that the compact complex hyperbolic surface $S = \mathbb{B}_{\mathbb{C}}^2/\Pi$ is smooth and has topological Euler characteristic $e(S) = 63$. Moreover, $\text{Aut}(S)$ contains an involution. Let Γ be the preimage of a subgroup $C_2 \leq \text{Aut}(S)$ in the normalizer $N_{PU(2,1)}(\Pi)$. Then Γ is a uniform lattice in $PU(2, 1)$, Γ contains an element of order 2, and*

$$\chi(\Gamma) = \frac{63}{2} \notin \mathbb{Z}.$$

Corollary 1.3 (Negative realization result for a uniform lattice with involution). *Let Γ be the uniform lattice produced in Theorem 1.2. Then there is no closed manifold M whose universal cover is $\mathbb{Q}$ -acyclic and such that $\pi_1(M) \cong \Gamma$.*

Proof of Corollary 1.3. If such an M existed, Theorem 1.1 would give $\chi(\Gamma) = \chi(M) \in \mathbb{Z}$. But Theorem 1.2 gives $\chi(\Gamma) = 63/2 \notin \mathbb{Z}$. Contradiction. $\square$

2 From a $\mathbb{Q}$ -acyclic universal cover to an integer Euler characteristic

2.1 Cellular chain complexes as free $\mathbb{Q}\Gamma$ -modules

Let M be a finite connected CW complex with $\pi_1(M) \cong \Gamma$. Fix a universal covering map $p : \widetilde{M} \rightarrow M$, and let Γ act on $\widetilde{M}$ by deck transformations.

Lemma 2.1 (Free module structure). *For each k , the cellular chain module $C_k(\widetilde{M}; \mathbb{Q})$ is a finitely generated free left $\mathbb{Q}\Gamma$ -module. More precisely, if c_k denotes the number of k -cells of M , then*

$$C_k(\widetilde{M}; \mathbb{Q}) \cong (\mathbb{Q}\Gamma)^{c_k}$$

as left $\mathbb{Q}\Gamma$ -modules.

Proof. Choose one lift of each k -cell of M to a k -cell in $\widetilde{M}$. The Γ -translates of these chosen lifts give a Γ -equivariant bijection between k -cells of $\widetilde{M}$ and $\Gamma \times \{1, \dots, c_k\}$. Thus $C_k(\widetilde{M}; \mathbb{Q})$, the $\mathbb{Q}$ -vector space spanned by k -cells, is isomorphic as a left $\mathbb{Q}\Gamma$ -module to the direct sum of c_k copies of $\mathbb{Q}\Gamma$. $\square$

2.2 A finite free resolution when the universal cover is $\mathbb{Q}$ -acyclic

Let $\mathbb{Q}$ denote the trivial $\mathbb{Q}\Gamma$ -module (i.e. Γ acts trivially).

Lemma 2.2 (Augmented chain complex). *Assume $\widetilde{M}$ is $\mathbb{Q}$ -acyclic. Then the augmented cellular chain complex*

$$0 \rightarrow C_n(\widetilde{M}; \mathbb{Q}) \xrightarrow{\partial_n} \dots \xrightarrow{\partial_2} C_1(\widetilde{M}; \mathbb{Q}) \xrightarrow{\partial_1} C_0(\widetilde{M}; \mathbb{Q}) \xrightarrow{\varepsilon} \mathbb{Q} \rightarrow 0$$

is exact, where ε is the augmentation map sending each 0-cell to $1 \in \mathbb{Q}$. Hence this is a finite free $\mathbb{Q}\Gamma$ -resolution of $\mathbb{Q}$.

Proof. Exactness of the augmented cellular chain complex is equivalent to vanishing of reduced homology $\widetilde{H}_*(\widetilde{M}; \mathbb{Q})$, which is the $\mathbb{Q}$ -acyclicity hypothesis. By Lemma 2.1, each $C_k(\widetilde{M}; \mathbb{Q})$ is a finitely generated free $\mathbb{Q}\Gamma$ -module. $\square$

2.3 Group Euler characteristic over $\mathbb{Q}$ and identification with $\chi(M)$

Definition 2.3 (Rank over $\mathbb{Q}\Gamma$). If $F \cong (\mathbb{Q}\Gamma)^r$ is a finitely generated free $\mathbb{Q}\Gamma$ -module, define $\text{rk}_{\mathbb{Q}\Gamma}(F) = r$. If P is finitely generated projective over $\mathbb{Q}\Gamma$, define $\text{rk}_{\mathbb{Q}\Gamma}(P)$ as the rank obtained after splitting P as a direct summand of a free module; equivalently, using the Kaplansky trace for group rings over fields.

Definition 2.4 (Euler characteristic of a group of type FP over $\mathbb{Q}$). If Γ admits a finite projective resolution $P_\bullet \rightarrow \mathbb{Q}$ by finitely generated projective $\mathbb{Q}\Gamma$ -modules, define

$$\chi(\Gamma) = \sum_{i \geq 0} (-1)^i \text{rk}_{\mathbb{Q}\Gamma}(P_i).$$

Standard homological algebra shows this is independent of the chosen finite projective resolution.

Lemma 2.5 (Integrality and equality with $\chi(M)$). *Under the hypotheses of Theorem 1.1, the resolution in Lemma 2.2 yields*

$$\chi(\Gamma) = \sum_{k=0}^n (-1)^k \text{rk}_{\mathbb{Q}\Gamma}(C_k(\widetilde{M}; \mathbb{Q})) = \sum_{k=0}^n (-1)^k c_k = \chi(M) \in \mathbb{Z}.$$

Proof. By Lemma 2.1, $\text{rk}_{\mathbb{Q}\Gamma}(C_k(\widetilde{M}; \mathbb{Q})) = c_k$. The alternating sum $\sum (-1)^k c_k$ is the definition of the Euler characteristic $\chi(M)$ of a finite CW complex. $\square$

Proof of Theorem 1.1. Combine Lemma 2.2 and Lemma 2.5. $\square$

3 Virtually torsion-free groups: index scaling and even-index under an involution

Uniform lattices in semisimple Lie groups are finitely generated linear groups; by Selberg's lemma they are virtually torsion-free. For Euler characteristic computations it is standard to use torsion-free finite-index subgroups.

3.1 Index scaling for Euler characteristic

Lemma 3.1 (Restriction of scalars scales rank). *Let $H \leq \Gamma$ be a subgroup of finite index d . Then $\mathbb{Q}\Gamma$ is a free left $\mathbb{Q}H$ -module of rank d . Consequently, for any finitely generated free $\mathbb{Q}\Gamma$ -module $F \cong (\mathbb{Q}\Gamma)^r$, viewed as a $\mathbb{Q}H$ -module by restriction, one has*

$$\mathrm{rk}_{\mathbb{Q}H}(F) = d \cdot r.$$

Proof. Choose a set of left coset representatives $g_1, \dots, g_d$ for $H \backslash \Gamma$. Then $\mathbb{Q}\Gamma = \bigoplus_{j=1}^d \mathbb{Q}H \cdot g_j$ as left $\mathbb{Q}H$ -modules, so $\mathbb{Q}\Gamma \cong (\mathbb{Q}H)^d$. Tensoring with $(\cdot)^r$ gives the claimed scaling for free modules. $\square$

Proposition 3.2 (Multiplicativity of Euler characteristic under finite index). *Let Γ be a group of type FP over $\mathbb{Q}$ and let $H \leq \Gamma$ have finite index d . Then H is also of type FP over $\mathbb{Q}$ and*

$$\chi(H) = d \cdot \chi(\Gamma).$$

Equivalently, $\chi(\Gamma) = \chi(H)/[\Gamma : H]$.

Proof. Let $P_\bullet \rightarrow \mathbb{Q}$ be a finite projective $\mathbb{Q}\Gamma$ -resolution of $\mathbb{Q}$. Restrict scalars to $\mathbb{Q}H$: each P_i remains finitely generated projective over $\mathbb{Q}H$, and the complex remains exact. Hence it is a finite projective $\mathbb{Q}H$ -resolution. Moreover, ranks scale by Lemma 3.1 (for projectives, use additivity after embedding as summands of frees). Thus

$$\chi(H) = \sum_i (-1)^i \mathrm{rk}_{\mathbb{Q}H}(P_i) = \sum_i (-1)^i d \mathrm{rk}_{\mathbb{Q}\Gamma}(P_i) = d \chi(\Gamma).$$

$\square$

3.2 Even index of torsion-free subgroups under 2-torsion

Lemma 3.3 (Finite subgroup order divides the index). *Let Γ be a group, let $H \leq \Gamma$ be torsion-free of finite index, and let $J \leq \Gamma$ be a finite subgroup. Then $|J|$ divides $[\Gamma : H]$. In particular, if Γ contains an involution, then every torsion-free finite-index subgroup has even index.*

Proof. Let Γ act on the finite set $X = \Gamma/H$ of left cosets by left multiplication. Restrict to the action of J . We claim the action of J on X is free. Indeed, if $j \in J$ fixes a coset gH , then $jgH = gH$, so $g^{-1}jg \in H$. But $g^{-1}jg$ has finite order (same as j) and lies in H ; since H is torsion-free, this forces $j = 1$. Thus every orbit of J on X has size $|J|$, so $|J|$ divides $|X| = [\Gamma : H]$. $\square$

4 Complex hyperbolic surfaces, normalizers, and half-integral Euler characteristics

4.1 Uniform lattices in $PU(2, 1)$ and compact ball quotients

Let $\mathbb{B}_{\mathbb{C}}^2$ denote the complex two-ball with its standard complex hyperbolic metric. Its full group of holomorphic isometries is $PU(2, 1)$.

Definition 4.1 (Uniform lattice). A subgroup $\Gamma < PU(2, 1)$ is a *uniform lattice* if it is discrete and the quotient orbifold $\mathbb{B}_{\mathbb{C}}^2/\Gamma$ is compact. If Γ is torsion-free, then $\mathbb{B}_{\mathbb{C}}^2/\Gamma$ is a smooth compact manifold (indeed a complex surface).

Lemma 4.2 (Euler characteristic of a compact aspherical manifold equals group Euler characteristic). *Let N be a closed aspherical manifold with $\pi_1(N) = H$. Then $BH \simeq N$, and hence $\chi(H) = \chi(N)$ (for the usual group Euler characteristic defined via a finite CW model of BH).*

Proof. Asphericity means $\tilde{N}$ is contractible, so N is a $K(H, 1)$. Thus $BH \simeq N$ and Euler characteristics agree. $\square$

Lemma 4.3 (Lifting automorphisms to the universal cover). *Let $\Pi < PU(2, 1)$ be torsion-free cocompact and let $S = \mathbb{B}_{\mathbb{C}}^2/\Pi$. Every holomorphic automorphism $f \in \text{Aut}(S)$ lifts to a holomorphic isometry $\tilde{f} \in PU(2, 1)$ of $\mathbb{B}_{\mathbb{C}}^2$ such that*

$$\tilde{f} \Pi \tilde{f}^{-1} = \Pi,$$

i.e. $\tilde{f}$ lies in the normalizer $N_{PU(2,1)}(\Pi)$. Moreover, different lifts differ by right multiplication by an element of Π . Consequently there is a short exact sequence

$$1 \longrightarrow \Pi \longrightarrow N_{PU(2,1)}(\Pi) \longrightarrow \text{Aut}(S) \longrightarrow 1.$$

Proof. Since $\mathbb{B}_{\mathbb{C}}^2$ is simply connected, $p : \mathbb{B}_{\mathbb{C}}^2 \rightarrow S$ is the universal covering map. Given $f \in \text{Aut}(S)$ and a basepoint $\tilde{x} \in \mathbb{B}_{\mathbb{C}}^2$, choose $\tilde{f}(\tilde{x})$ to be any point in the fiber over $f(p(\tilde{x}))$. By the lifting criterion for covering spaces, f admits a unique lift $\tilde{f}$ sending $\tilde{x}$ to that chosen point. Because f is holomorphic and p is a local biholomorphism, $\tilde{f}$ is holomorphic. It follows $\tilde{f}$ is a holomorphic isometry of $\mathbb{B}_{\mathbb{C}}^2$, hence $\tilde{f} \in PU(2, 1)$.

For $\pi \in \Pi$ a deck transformation, $p \circ \pi = p$. Because $p \circ \tilde{f} = f \circ p$, we have

$$p \circ (\tilde{f} \pi \tilde{f}^{-1}) = f \circ p \circ \pi \circ \tilde{f}^{-1} = f \circ p \circ \tilde{f}^{-1} = p,$$

so $\tilde{f} \pi \tilde{f}^{-1}$ is a deck transformation, hence lies in Π . Thus $\tilde{f} \Pi \tilde{f}^{-1} \subseteq \Pi$ and by symmetry equality holds.

If $\tilde{f}_1, \tilde{f}_2$ are two lifts of f , then $\tilde{f}_2^{-1} \tilde{f}_1$ is a deck transformation, hence lies in Π . This gives the exact sequence: the map $N_{PU(2,1)}(\Pi) \rightarrow \text{Aut}(S)$ sends a normalizer element to the induced automorphism on S , its kernel is precisely Π , and surjectivity follows from existence of lifts. $\square$

4.2 Odd Euler characteristic plus an involution yields a half-integral Euler characteristic lattice

Proposition 4.4 (Odd $e(S)$ and an involution force half-integral Euler characteristic). *Let $\Pi < PU(2, 1)$ be torsion-free cocompact and let $S = \mathbb{B}_{\mathbb{C}}^2/\Pi$. Assume:*

1. $e(S) = \chi(S)$ is odd, and
2. $\text{Aut}(S)$ contains an involution (equivalently, $|\text{Aut}(S)|$ is even).

Let $C_2 \leq \text{Aut}(S)$ be a subgroup of order 2 and let Γ be its full preimage in $N_{PU(2,1)}(\Pi)$. Then:

1. Γ is a uniform lattice in $PU(2, 1)$ containing an element of order 2,
2. $[\Gamma : \Pi] = 2$, and
3. $\chi(\Gamma) = \chi(\Pi)/2 = e(S)/2 \notin \mathbb{Z}$.

Proof. By Lemma 4.3 we have $1 \rightarrow \Pi \rightarrow N_{PU(2,1)}(\Pi) \rightarrow \text{Aut}(S) \rightarrow 1$. Let Γ be the preimage of C_2 ; then $\Pi \trianglelefteq \Gamma$ and $\Gamma/\Pi \cong C_2$, hence $[\Gamma : \Pi] = 2$. Any lift of the nontrivial element of C_2 is an element of order 2 in Γ (since its square lies in Π and projects to the identity, and torsion-freeness of Π forces the square to be the identity), so Γ contains an involution. Discreteness and cocompactness of Γ follow because Π is discrete cocompact and Γ contains Π as finite index subgroup.

Since S is aspherical (its universal cover is $\mathbb{B}_{\mathbb{C}}^2$, contractible), Lemma 4.2 gives $\chi(\Pi) = \chi(S) = e(S)$. Finally Proposition 3.2 with $H = \Pi$ yields

$$\chi(\Gamma) = \frac{\chi(\Pi)}{[\Gamma : \Pi]} = \frac{e(S)}{2}.$$

If $e(S)$ is odd then this is not an integer. □

5 Existence of an odd-Euler ball quotient with an involution

Theorem 5.1 (Existence of a smooth compact complex hyperbolic surface with $e = 63$ and even automorphism group). *There exists a smooth compact arithmetic complex hyperbolic surface S (a torsion-free ball quotient $\mathbb{B}_{\mathbb{C}}^2/\Pi$) with topological Euler characteristic $e(S) = 63$ and with finite automorphism group $\text{Aut}(S)$ of even order.*

Remark 5.2. In known constructions one can take $\text{Aut}(S) \cong \text{PSU}(3, \mathbb{F}_3) \times (\mathbb{Z}/3\mathbb{Z})$, whose order is even, hence it contains an involution by Cauchy's theorem.

Proof of Theorem 1.2. Let S be as in Theorem 5.1, so $S = \mathbb{B}_{\mathbb{C}}^2/\Pi$ for torsion-free cocompact $\Pi < PU(2,1)$, and $e(S) = 63$ is odd. Since $|\text{Aut}(S)|$ is even, $\text{Aut}(S)$ contains a subgroup C_2 . Apply Proposition 4.4 to obtain the over-lattice Γ with $[\Gamma : \Pi] = 2$ and

$$\chi(\Gamma) = \frac{e(S)}{2} = \frac{63}{2} \notin \mathbb{Z},$$

and Γ contains an involution. This proves Theorem 1.2. □

6 Conclusion: obstruction to $\mathbb{Q}$ -acyclic universal covers

Proof of Theorem 1.1 + Theorem 1.2 $\Rightarrow$ Corollary 1.3. Theorem 1.1 asserts that any closed manifold M with $\widetilde{M}$ $\mathbb{Q}$ -acyclic and $\pi_1(M) \cong \Gamma$ forces $\chi(\Gamma) \in \mathbb{Z}$. Theorem 1.2 provides a uniform lattice Γ with an involution and $\chi(\Gamma) = 63/2 \notin \mathbb{Z}$. Therefore no such M exists for this Γ . □

A Brown's p -fractional-part theory and the $p = 2$ involution reduction

This appendix records the full machinery developed in the intermediate reductions. It is not needed to build the explicit counterexample in the main text, but it provides a general framework for controlling the 2-adic (and p -adic) fractional parts of Euler characteristics of groups with torsion, and it explains why involutions naturally govern the 2-fractional part.

A.1 Equivariant Euler characteristic of a Γ -complex

Definition A.1 (Equivariant Euler characteristic). Let a group Γ act cellularly on a finite Γ -CW complex K with finitely many Γ -orbits of cells, and assume each cell stabilizer Γ_σ has a well-defined Euler characteristic $\chi(\Gamma_\sigma)$ (e.g. is of type FP over $\mathbb{Q}$). Let Σ be a set of representatives for Γ -orbits of open cells. Define

$$\chi_\Gamma(K) = \sum_{\sigma \in \Sigma} (-1)^{\dim \sigma} \chi(\Gamma_\sigma).$$

Lemma A.2 (Orbit-sum formula). *With the hypotheses above, $\chi_\Gamma(K)$ is well-defined (independent of orbit representatives) and is additive over Γ -cell decompositions.*

A.2 The Brown poset of p -subgroups

Fix a prime p .

Definition A.3 (Poset $S_p(\Gamma)$). Let $S_p(\Gamma)$ be the poset (by inclusion) of nontrivial finite p -subgroups of Γ . Let $K(S_p(\Gamma))$ denote its order complex: its k -simplices are strict chains $P_0 < P_1 < \cdots < P_k$ in $S_p(\Gamma)$.

Lemma A.4 (Stabilizers are chain normalizers). *For a chain $\sigma = (P_0 < \cdots < P_k)$, its stabilizer in Γ equals the normalizer of the chain:*

$$\Gamma_\sigma = N_\Gamma(P_0 < \cdots < P_k) := \{\gamma \in \Gamma : \gamma P_i \gamma^{-1} = P_i \ \forall i\}.$$

A.3 Brown's p -fractional-part theorem

Definition A.5 (p -integral numbers and $\sim^p$). A rational number $a \in \mathbb{Q}$ is p -integral if its denominator is not divisible by p , i.e. $a \in \mathbb{Z}_{(p)}$. Write $x \sim^p y$ if $x - y \in \mathbb{Z}_{(p)}$.

Theorem A.6 (Brown: p -fractional part via $S_p(\Gamma)$). *Assume Γ is of finite homological type (e.g. type FP over $\mathbb{Q}$) and satisfies:*

1. Γ has finitely many conjugacy classes of finite p -subgroups, and
2. for each finite p -subgroup P , the normalizer $N_\Gamma(P)$ is of finite homological type,

so that the equivariant Euler characteristic $\chi_\Gamma(S_p(\Gamma)) := \chi_\Gamma(K(S_p(\Gamma)))$ is defined. Then

$$\chi(\Gamma) \sim^p \chi_\Gamma(S_p(\Gamma)).$$

A.4 Compression of the chain sum to a subgroup sum

Proposition A.7 (Compression). *Under the hypotheses of Theorem A.6, there exist rational coefficients $\mu_p(P)$ depending only on the finite p -group P (up to isomorphism) such that*

$$\chi_\Gamma(S_p(\Gamma)) = \sum_{[P] \neq 1} \mu_p(P) \chi(N_\Gamma(P)),$$

where $[P]$ runs over conjugacy classes of nontrivial finite p -subgroups. Moreover, for $P \cong C_p$ one has $\mu_p(C_p) = 1$.

Proof. Start from the orbit-sum definition applied to $K(S_p(\Gamma))$, i.e.

$$\chi_\Gamma(S_p(\Gamma)) = \sum_{[\sigma]} (-1)^{\dim \sigma} \chi(\Gamma_\sigma) = \sum_{[\sigma]} (-1)^{\dim \sigma} \chi(N_\Gamma(\sigma)).$$

Group the Γ -orbits of chains by the conjugacy class of their top element $P = \max(\sigma)$. For fixed P , chains ending at P correspond to chains in the poset $\mathcal{S}(P)$ of nontrivial proper subgroups of P , and stabilizers are precisely chain-normalizers inside $N_\Gamma(P)$. This yields a factorization of the inner alternating sum as a coefficient depending only on $\mathcal{S}(P)$ times $\chi(N_\Gamma(P))$, giving the desired coefficients $\mu_p(P)$. When $P \cong C_p$ there are no nontrivial proper subgroups, so only the 0-simplex contributes, giving coefficient 1. $\square$

A.5 The special role of involutions: evenness of coefficients for $p = 2$

We now specialize to $p = 2$.

Theorem A.8 (Evenness of coefficients for 2-groups). *Let P be a finite 2-group. Then $\mu_2(C_2) = 1$, and if $|P| > 2$ then $\mu_2(P) \in 2\mathbb{Z} \subset \mathbb{Z}_{(2)}$.*

Proof. One may realize $\mu_2(P)$ (up to a global sign convention fixed by $\mu_2(C_2) = 1$) as the reduced Euler characteristic of the order complex of the poset of nontrivial proper subgroups of P . Standard results in the topology of subgroup complexes imply:

- If P is not elementary abelian, then this poset is contractible, hence reduced Euler characteristic is 0.
- If $P \cong (\mathbb{Z}/2)^r$ with $r \geq 2$, then the poset is the lattice of proper nonzero subspaces of $\mathbb{F}_2^r$, whose reduced Euler characteristic equals the Möbius function value $(-1)^{r-1} 2^{\binom{r}{2}}$, which is even for $r \geq 2$.

This yields the claim. $\square$

A.6 Involution-normalizer reduction modulo 2-integrals

Corollary A.9 (Only order-2 subgroups contribute mod $\mathbb{Z}_{(2)}$). *Under the hypotheses of Theorem A.6 with $p = 2$,*

$$\chi(\Gamma) \equiv \sum_{[C_2]} \chi(N_\Gamma(C_2)) \pmod{\mathbb{Z}_{(2)}},$$

where the sum ranges over conjugacy classes of subgroups of order 2 in Γ .

Proof. By Theorem A.6 and Proposition A.7,

$$\chi(\Gamma) \sim^2 \sum_{[P] \neq 1} \mu_2(P) \chi(N_\Gamma(P)).$$

By Theorem A.8, the terms with $|P| > 2$ lie in $\mathbb{Z}_{(2)}$, hence vanish modulo $\mathbb{Z}_{(2)}$. The surviving terms correspond to $P \cong C_2$ with coefficient 1. $\square$

B Equal-rank dichotomy for compact locally symmetric manifolds

This appendix records the equal-rank/unequal-rank vanishing phenomenon for Euler characteristics of compact locally symmetric manifolds, as used in the broader (unfinished) attempt to prove universal statements about lattices with torsion.

B.1 Locally symmetric manifolds from torsion-free lattices

Let G be a connected semisimple real Lie group with maximal compact subgroup K and symmetric space $X = G/K$ (contractible). If $\Gamma' \leq G$ is torsion-free cocompact, then $Y = \Gamma' \backslash X$ is a closed aspherical manifold.

B.2 Olbrich's L^2 -invariants and Euler characteristic

Theorem B.1 (Olbrich: L^2 -Betti numbers and Euler characteristic). *Let $Y = \Gamma' \backslash X$ be as above and define the fundamental rank*

$$m := \mathrm{rk}_{\mathbb{C}}(G) - \mathrm{rk}_{\mathbb{C}}(K).$$

Then $b_p^{(2)}(Y) \neq 0$ iff $m = 0$ and $p = \dim(Y)/2$. In particular, if $m > 0$ then $\chi(Y) = 0$. If $m = 0$, then

$$(-1)^{\dim(Y)/2} \chi(Y) = \frac{\mathrm{vol}(Y)}{\mathrm{vol}(X_d)} \chi(X_d),$$

where X_d is the compact dual symmetric space.

Remark B.2. Combined with Lemma 3.3 and Proposition 3.2, this yields a sharp dichotomy for $\chi(\Gamma)$ when Γ has torsion and Γ' is torsion-free finite index: either $\chi(\Gamma) = 0$ in the unequal-rank case, or else integrality constraints in the equal-rank case reduce to parity/2-adic questions.